

This was some preparatory work for expanding the cybersecurity lab on my computer. Today I was focused on installing part of Metasploitable 3 which was the Windows Server 2008 variant. There is also a variant based on Ubuntu but I will install that later on.

First I started out with installing Vagrant which would be a vital part in making sure this works. I went through Vagrant's small install process which took a few minutes and ended with me needing to restart my computer for its changes to take effect.

After restarting my computer I opened up command prompt and typed in “vagrant plugin install vagrant-reload” to install one of the plugins needed. After that I typed in “vagrant plugin install vagrant-vbguest” into command prompt, which this installed two plugins.

```
C:\Users\Brandon>vagrant plugin install vagrant-reload
Installing the 'vagrant-reload' plugin. This can take a few minutes...
Fetching vagrant-reload-0.0.1.gem
Installed the plugin 'vagrant-reload (0.0.1)'!
```

```
C:\Users\Brandon>vagrant plugin install vagrant-vbguest
Installing the 'vagrant-vbguest' plugin. This can take a few minutes...
Fetching micromachine-3.0.0.gem
Fetching vagrant-vbguest-0.32.0.gem
Installed the plugin 'vagrant-vbguest (0.32.0)'!
```

After those two commands I typed “vagrant box add rapid7/metasploitable3-win2k8” into the command prompt, and it loaded for a bit before showing me a small menu with three options to select for me to let it know what virtualization it will be using. I input “1” into the command prompt for Virtualbox and it began to install files I need for Virtualbox.

```

C:\Users\Brandon>vagrant box add rapid7/metasploitable3-win2k8
==> box: Loading metadata for box 'rapid7/metasploitable3-win2k8'
      box: URL: https://vagrantcloud.com/api/v2/vagrant/rapid7/metasploitable3-win2k8
This box can work with multiple providers! The providers that it
can work with are listed below. Please review the list and choose
the provider you will be working with.

1) virtualbox
2) vmware
3) vmware_desktop

Enter your choice: 1
==> box: Adding box 'rapid7/metasploitable3-win2k8' (v0.1.0-weekly) for provider: virt
ualbox
      box: Downloading: https://vagrantcloud.com/rapid7/boxes/metasploitable3-win2k8/ver
sions/0.1.0-weekly/providers/virtualbox/unknown/vagrant.box
      box:
==> box: Successfully added box 'rapid7/metasploitable3-win2k8' (v0.1.0-weekly) for 'v
irtualbox'!

```

After that I typed in “cd .vagrant.d/boxes” which changed the directory I was currently in, to the new directory specified in the command. I then typed in the “dir” command which showed me everything in this directory.

```

C:\Users\Brandon\.vagrant.d\boxes>dir
Volume in drive C is Windows
Volume Serial Number is 0EBA-0155

Directory of C:\Users\Brandon\.vagrant.d\boxes

02/19/2026  11:40 AM    <DIR>          .
02/19/2026  11:33 AM    <DIR>          ..
02/19/2026  11:40 AM    <DIR>          rapid7-VAGRANTSLASH-metasploitable3-win2k8
               0 File(s)                0 bytes
               3 Dir(s)  143,809,789,952 bytes free

```

I went to go rename one of the files “rapid7-VAGRANTSHLASH-metasploitable3-win2k8” to “metasploitable3-win2k8” using the command “REN “rapid7-VAGRANTSHLASH-metasploitable3-win2k8” “metasploitable3-win2k8””

```

C:\Users\Brandon\.vagrant.d\boxes>REN "rapid7-VAGRANTSLASH-metasploitable3-win2k8" "metasploitable3-win2k8"

C:\Users\Brandon\.vagrant.d\boxes>dir
Volume in drive C is Windows
Volume Serial Number is 0EBA-0155

Directory of C:\Users\Brandon\.vagrant.d\boxes

02/19/2026  11:49 AM    <DIR>          .
02/19/2026  11:33 AM    <DIR>          ..
02/19/2026  11:40 AM    <DIR>          metasploitable3-win2k8
               0 File(s)                0 bytes
               3 Dir(s)  143,808,212,992 bytes free

```

I typed in “vagrant init metasploitable3-win2k8” into command prompt. This then gave me a ‘Vagrantfile’ placed in this directory.

```

C:\Users\Brandon\.vagrant.d\boxes>vagrant init metasploitable3-win2k8
A 'Vagrantfile' has been placed in this directory. You are now
ready to 'vagrant up' your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
'vagrantup.com' for more information on using Vagrant.

```

I then typed in “vagrant up” into command prompt (metasploitable 3 has some issues and this could have triggered an issue for me but it didn't really, something else happened, but I did fix the small bug in one of the files that could cause the issue-line 84 removing the ‘s’ from exists so it is in line with the code on line 75).

```

C:\Users\Brandon\.vagrant.d\boxes>vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Importing base box 'metasploitable3-win2k8'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'metasploitable3-win2k8' version '0.1.0-weekly' is up to
date...
==> default: Setting the name of the VM: boxes_default_1771520585092_49178
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
default: Adapter 1: nat
==> default: Forwarding ports...
default: 3389 (guest) => 3389 (host) (adapter 1)
default: 22 (guest) => 2222 (host) (adapter 1)
default: 5985 (guest) => 55985 (host) (adapter 1)
default: 5986 (guest) => 55986 (host) (adapter 1)
==> default: Running 'pre-boot' VM customizations...
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes...
default: WinRM address: 127.0.0.1:55985
default: WinRM username: vagrant
default: WinRM execution_time_limit: PT2H
default: WinRM transport: negotiate
==> default: Machine booted and ready!
==> default: Checking for guest additions in VM...

```

```
==> default: Machine booted and ready!
==> default: Checking for guest additions in VM...
default: The guest additions on this VM do not match the installed version of
default: VirtualBox! In most cases this is fine, but in rare cases it can
default: prevent things such as shared folders from working properly. If you see
default: shared folder errors, please make sure the guest additions within the
default: virtual machine match the version of VirtualBox you have installed on
default: your host and reload your VM.
default:
default: Guest Additions Version: 6.0.8
default: VirtualBox Version: 7.2
Vagrant can't use the requested machine because it is locked! This
means that another Vagrant process is currently reading or modifying
the machine. Please wait for that Vagrant process to end and try
again. Details about the machine are shown below:

Name: default
Provider: virtualbox
```

This came up but this may have been caused since I tabbed back into Virtualbox on my computer and the Windows Server 2008 VM was running already.

Next I shut the VM down and went to go adjust some settings on the VM, mainly giving the VM 2GB of memory and changing the network settings to the host-only adapter. I then renamed the VM to "Metasploitable3-win2k8".

I started up both the Metasploitable3-win2k8 VM and the Kali Linux VM so I could make sure that both of these boxes could communicate with each other. I logged into the administrator account on Metasploitable3-win2k8 and logged into my Kali Linux account. I opened up command prompt on the Metasploitable3-win2k8 box and typed in "ipconfig" in order to get the IP address of that box. On the Kali Linux machine I opened up the terminal and used the "ping" command along with the IP address from the Metasploitable3-win2k8 box and hit enter. Both boxes were able to communicate with each other.